



student@ubuntu: ~

Linux Fundamentals

Session 1

Linux Learning

What Will We Learn Today?



What is Linux?

OS, kernel & distributions



Linux history

UNIX, GNU and Linux



Kernel

What it manages



Why Linux matters

Servers, cloud, AI



Installation & WSL

Linux on Windows



Terminal, Shell & Bash

How you talk to Linux



GUI vs CLI

Two ways to work



Essential commands

Navigate & manage files

By the end of this session, you will be able to navigate and work inside a Linux environment.

What Is an Operating System?

DEFINITION

An operating system is system software that manages computer hardware and provides an environment for applications to run.

Examples

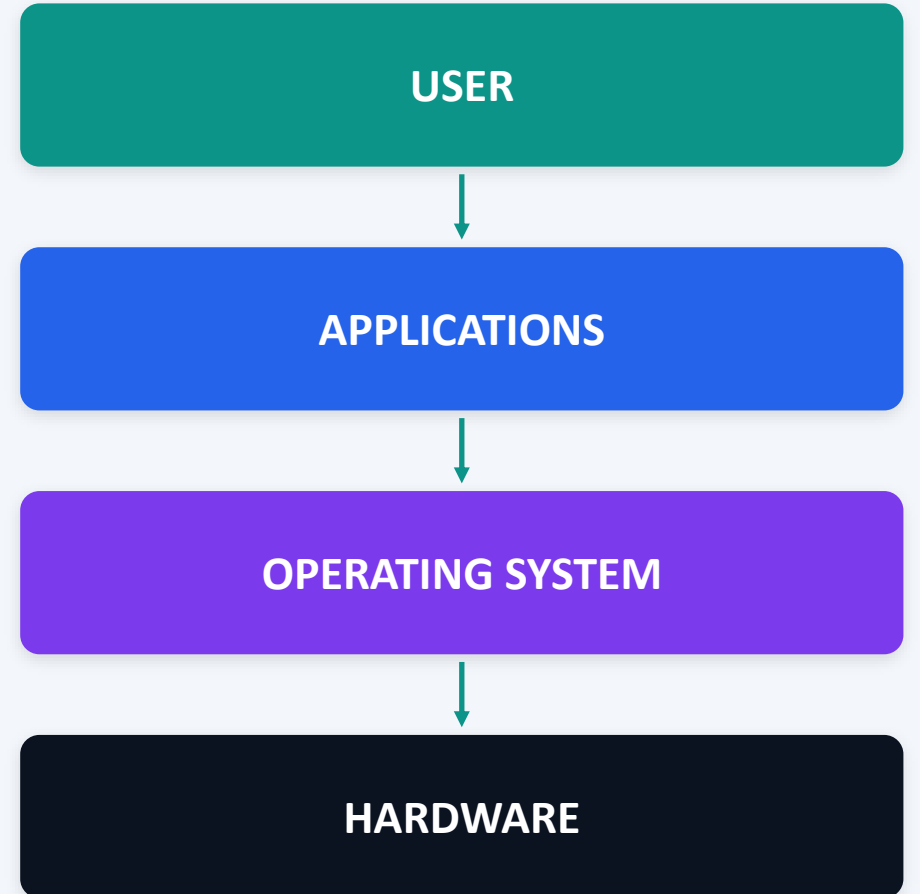
Windows

macOS

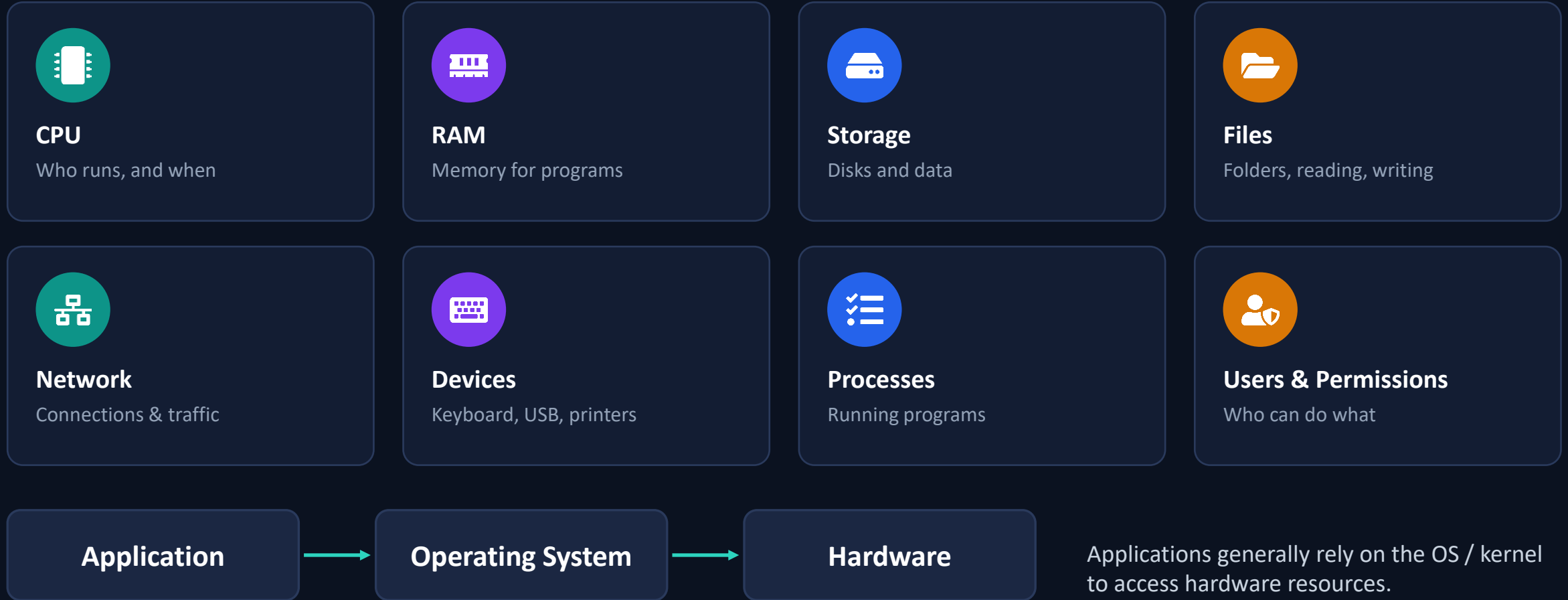
Linux distributions

Android

iOS



What Does an Operating System Actually Do?



What is Linux?

DEFINITION

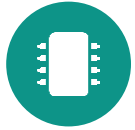
Linux is an open-source operating system kernel that forms the core of many operating systems used on servers, desktops, cloud platforms and embedded devices.

Linux = Kernel

A complete Linux operating system (a distribution) combines the Linux kernel with system utilities, libraries, package management and applications.



Linux ≠ Ubuntu



Linux

- The kernel
- Core of the system
- Manages hardware & resources



Ubuntu

- A Linux distribution
- Includes the Linux kernel
- Includes utilities & package manager
- Includes apps & system tools

Linux Kernel



Ubuntu

Debian

Fedora

Arch

Rocky Linux

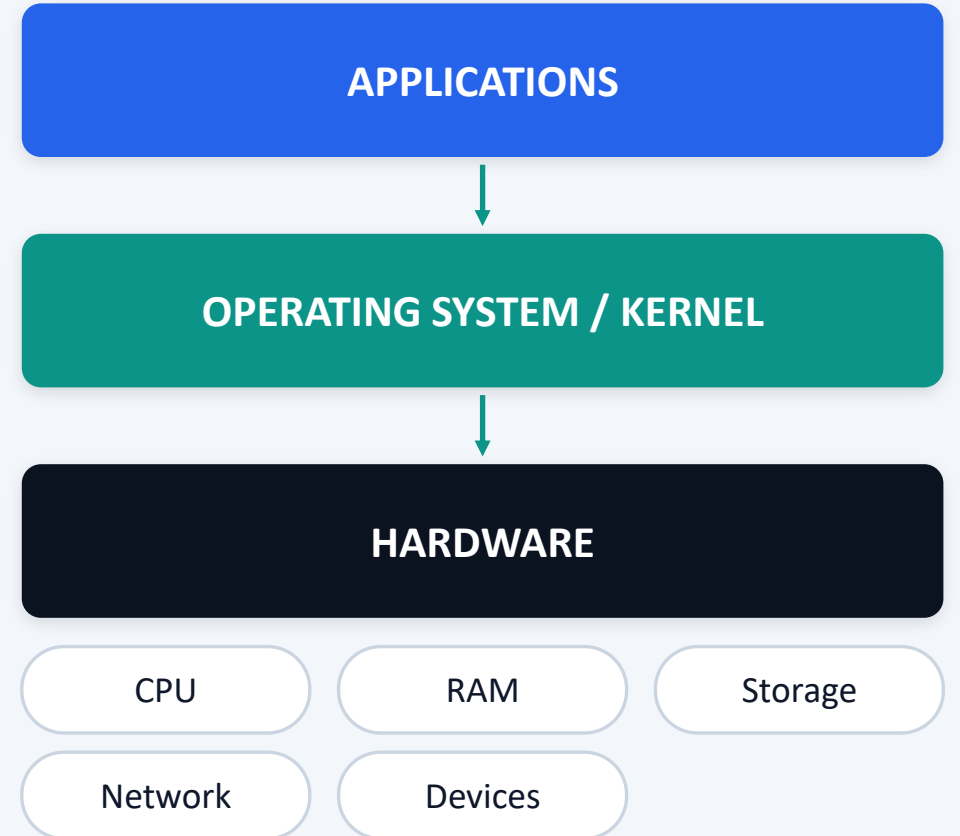
What is a Kernel?

DEFINITION

The kernel is the core component of an operating system that manages hardware resources and provides a bridge between applications and hardware.

Analogy

Kernel = the manager between software and hardware



Responsibilities of the Linux Kernel



Process Management

Schedules running programs



Memory Management

Gives programs their RAM



File System

Access to files & disks



Networking

Network communication



Device Drivers

Talks to devices



Security & Permissions

Controls access



Hardware Interaction

CPU, RAM, disks, USB

EXAMPLE

```
Python app
  ↓
Kernel
  ↓
RAM / CPU
```

Applications

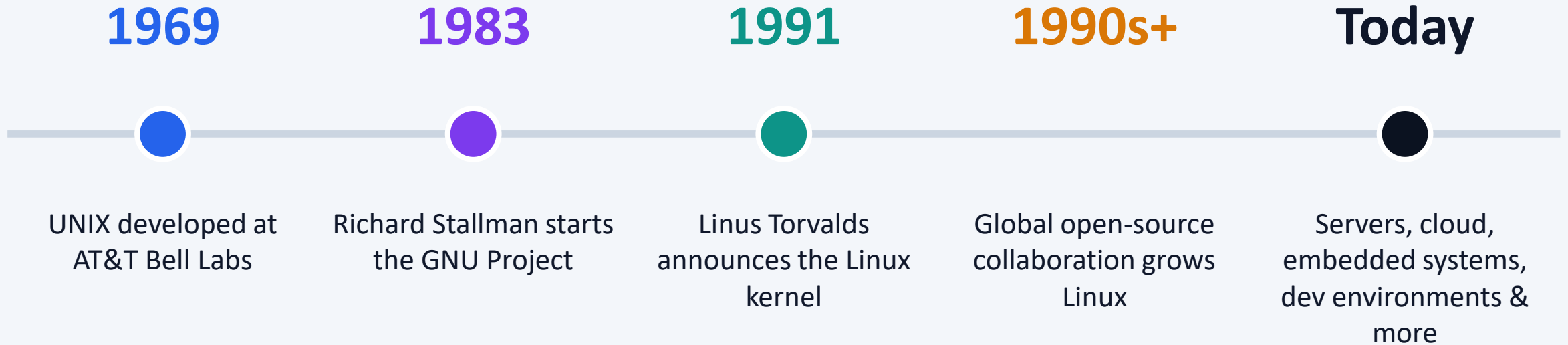


Linux Kernel



Hardware

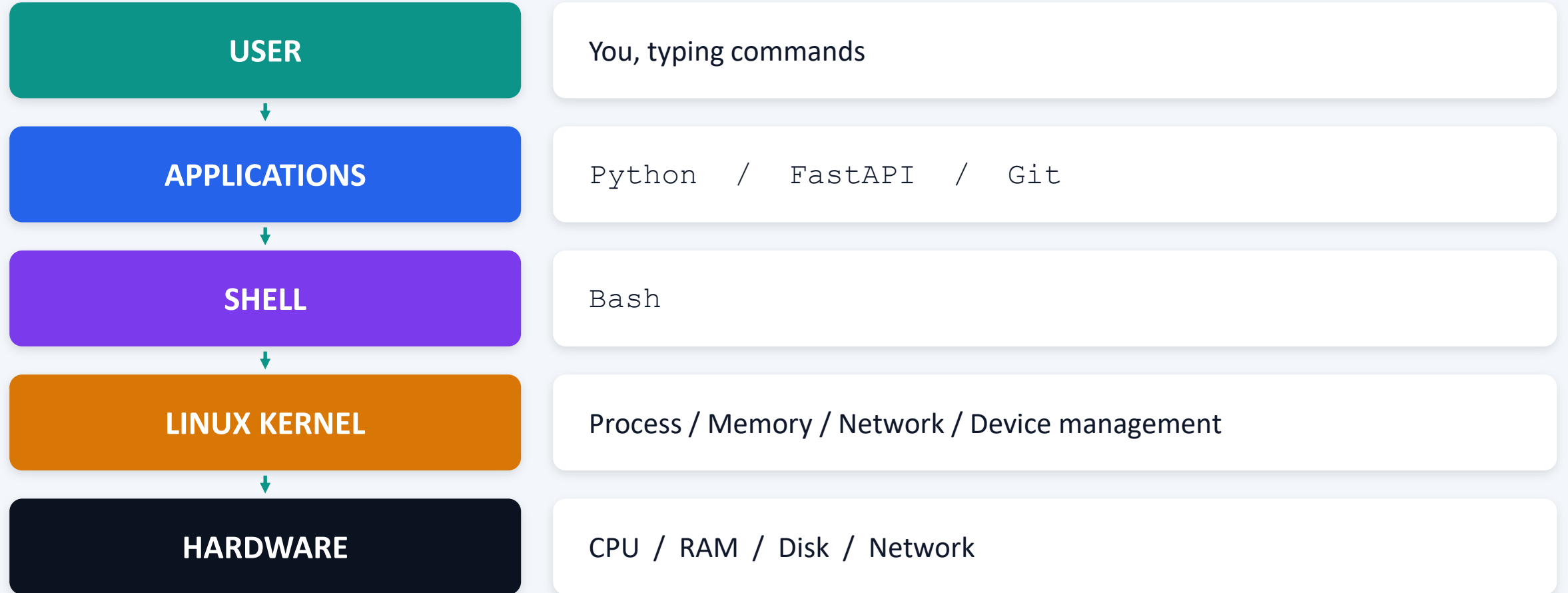
Linux History — From UNIX to Linux



How Did the Linux Ecosystem Evolve?



Linux at a High Level



Why Should Developers Learn Linux?



Servers

Where apps run



Cloud Computing

AWS, Azure, GCP



Backend Development

APIs and services



DevOps

Build, deploy, operate



Containers & Docker

Built on Linux features



AI / ML Infrastructure

Training & serving



Remote Management

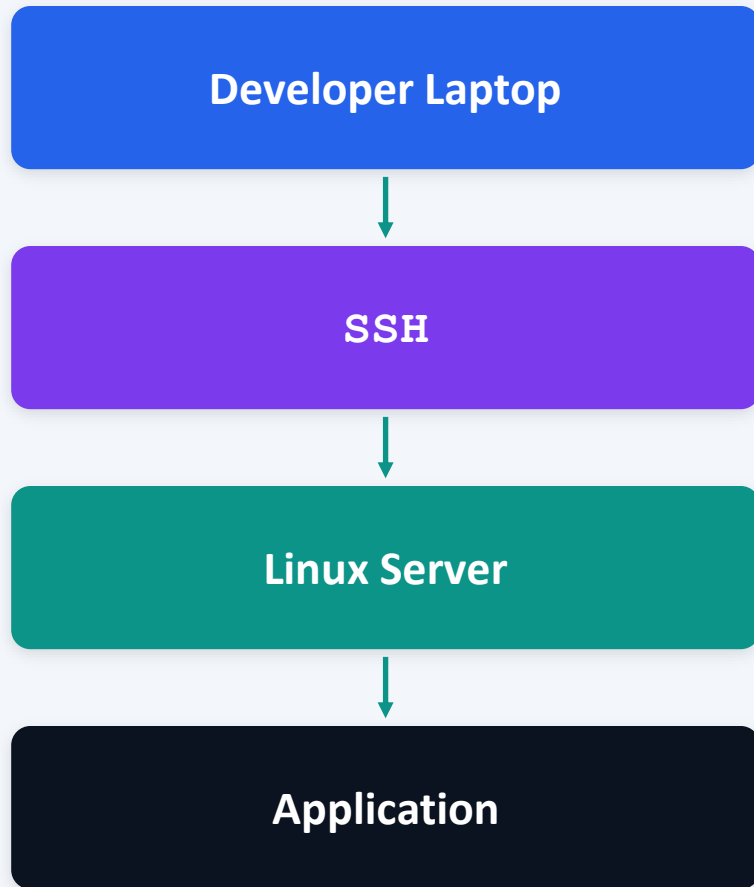
Work over SSH



Automation

Scripts save time

Why is Linux Important for Servers?



Command-line first

Servers are often managed through command-line tools.



No desktop needed

Remote servers may not have a graphical desktop.



Terminal skills are essential

Developers manage applications and infrastructure from the terminal.

How Can We Use Linux?



1. WSL

Linux running inside Windows

- Easy for Windows users
- No need to replace Windows
- Fast setup
- Excellent for learning the Linux CLI



2. Virtual Machine

A full Linux system running in a virtual machine on your computer

- Complete Linux environment
- Isolated from your main OS
- Uses extra RAM and disk



3. Dual Boot / Native

Linux installed on the computer, beside or instead of Windows

- Linux uses the hardware directly
- Restart to switch OS (dual boot)
- Best when Linux is your main OS

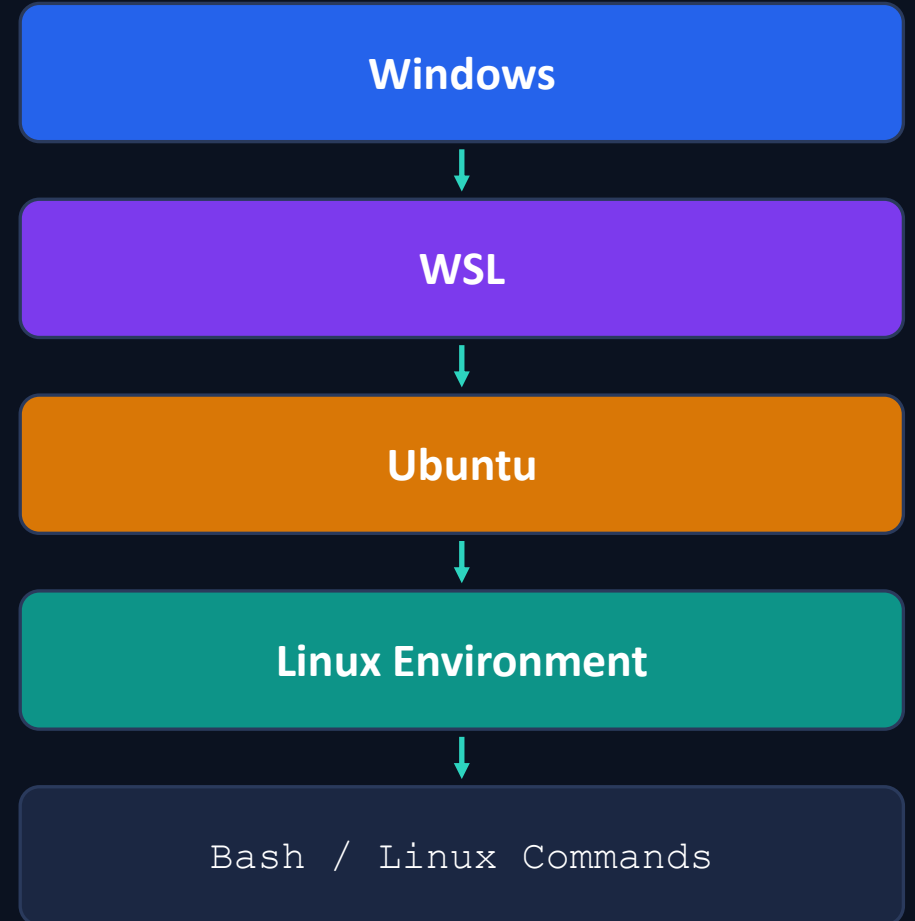
What is WSL?

DEFINITION

WSL (Windows Subsystem for Linux) allows users to run Linux distributions and Linux command-line environments directly on Windows.

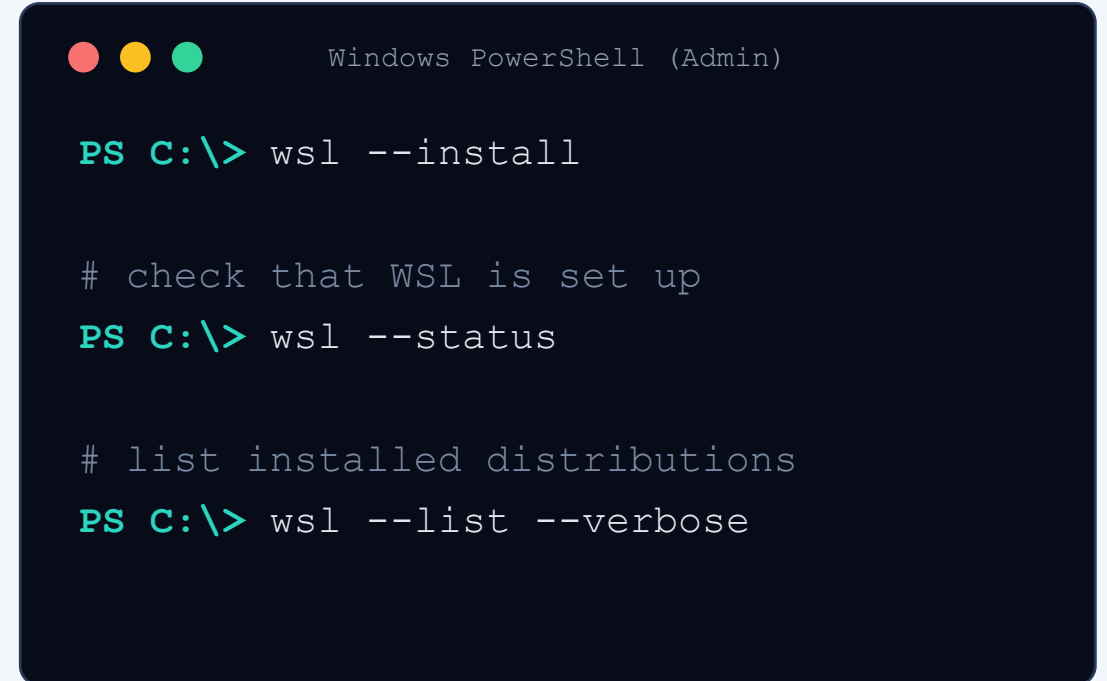
IMPORTANT

Windows Terminal is not Linux. It is an application that can host shells such as Bash running through WSL.



Installing Linux with WSL

- 1 Open PowerShell as Administrator
- 2 Run `wsl --install`
- 3 Restart if requested
- 4 Launch Ubuntu
- 5 Create your Linux username & password
- 6 Start using the terminal



```
Windows PowerShell (Admin)

PS C:\> wsl --install

# check that WSL is set up
PS C:\> wsl --status

# list installed distributions
PS C:\> wsl --list --verbose
```

Exact installation behaviour can vary slightly depending on Windows version and configuration.

What is a Terminal?

DEFINITION

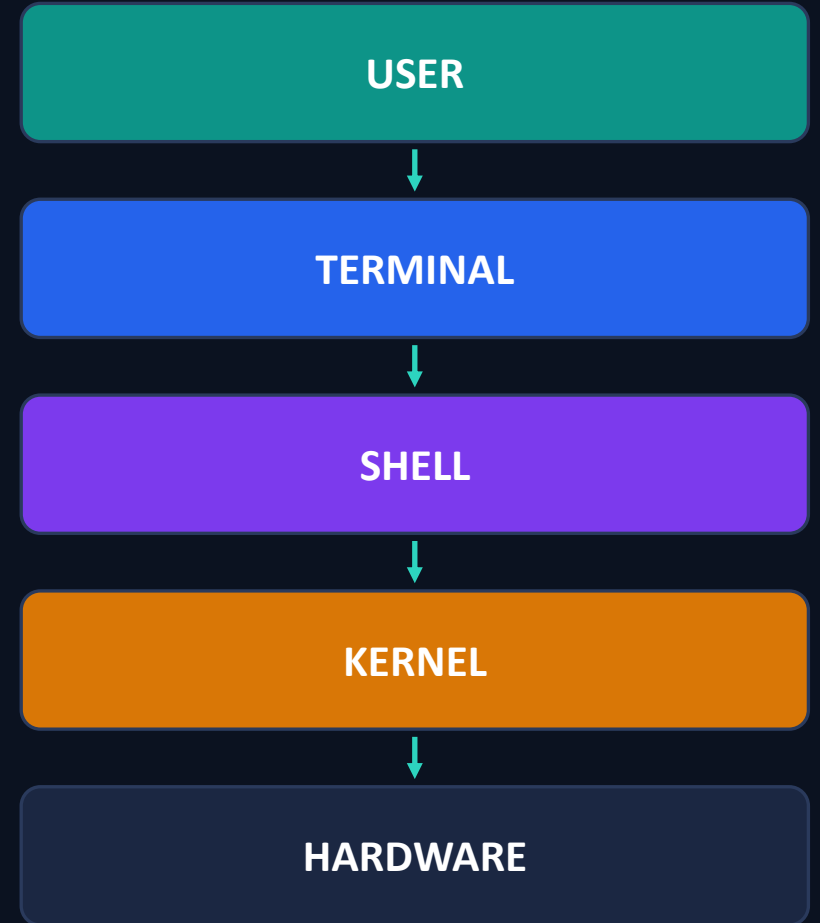
A terminal is a text-based interface through which a user interacts with a computer by entering commands and viewing their output.

Terminal

provides the interface

Shell

interprets the commands



Don't Confuse These 3



TERMINAL

- Interface / application
- Lets you interact with a shell



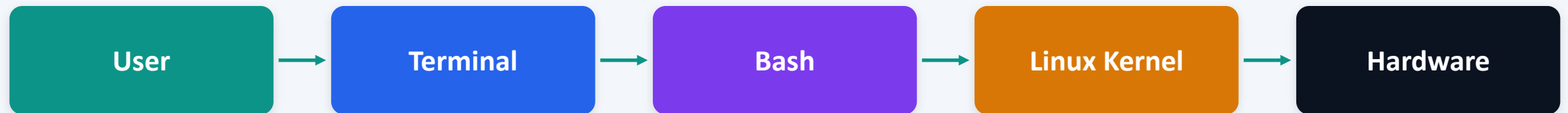
SHELL

- Command interpreter
- Executes / interprets your commands



BASH

- A specific shell
- Commonly used on Linux



GUI vs Command Line



- Mouse-driven
- Windows and icons
- Menus
- Visual
- Easy for beginners



- Commands and keyboard
- Text-based
- Great for automation
- Powerful
- Common on servers

GUI	CLI
Right click → New Folder	<code>mkdir project</code>
Copy the file (Ctrl+C, Ctrl+V)	<code>cp app.py backup.py</code>

Why Do Developers Use the Terminal?



On a remote Linux server, the terminal is often the primary way developers and administrators interact with the system.



Speed



Automation



Remote Servers



Scripting



DevOps



Cloud

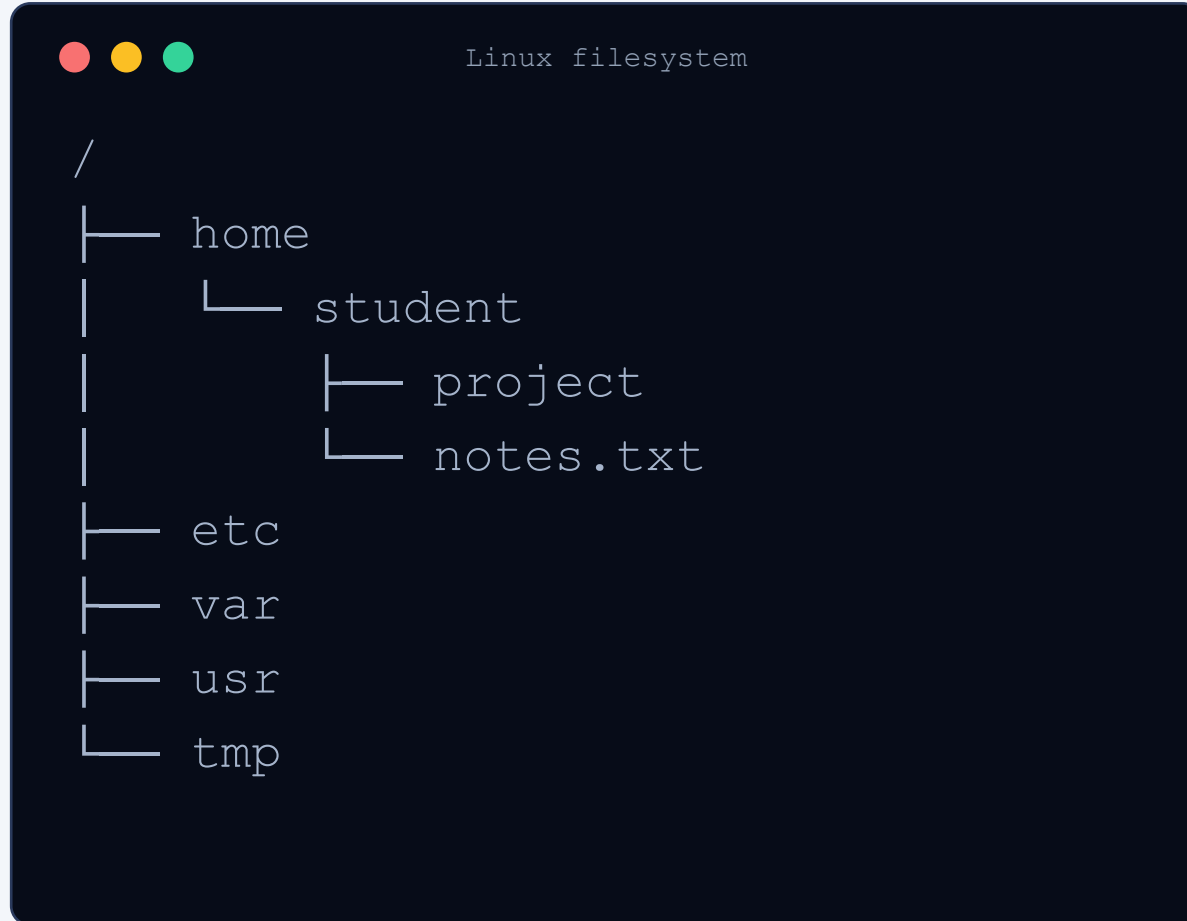


Containers



Server Management

Understanding the Linux Filesystem



Root directory

The top of the entire Linux filesystem.

/home/student

Home directory

An example of a user's own home directory.

How Linux Commands Work

`command` `[options]` `[arguments]`

`ls`

COMMAND

the program to run

`-l`

OPTION

changes how it behaves

`/home`

ARGUMENT

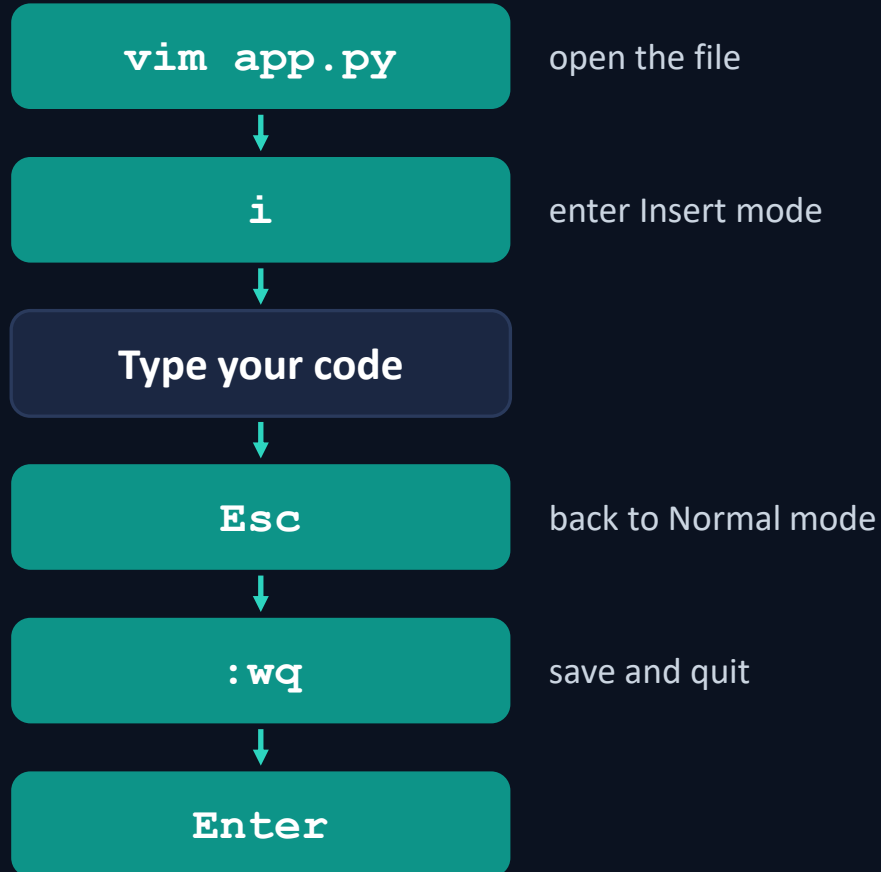
what to act on

Not every command needs options or arguments. Example: `pwd`

Essential Linux Commands — Session 1

Category	Command	Purpose	Example
Navigation	<code>pwd</code>	Show current directory	<code>pwd</code>
	<code>cd</code>	Change directory	<code>cd project</code>
	<code>ls</code>	List files and directories	<code>ls -la</code>
Directory / File Creation	<code>mkdir</code>	Create a directory	<code>mkdir project</code>
	<code>touch</code>	Create / update a file	<code>touch app.py</code>
File Editing	<code>vim</code>	Edit a text file	<code>vim app.py</code>
File Operations	<code>cp</code>	Copy files / directories	<code>cp app.py backup.py</code>
File Viewing	<code>cat</code>	Display file contents	<code>cat app.py</code>
Terminal Utilities	<code>clear</code>	Clear the visible screen	<code>clear</code>
	<code>history</code>	Show previous commands	<code>history</code>

Editing Files from the Terminal



Normal mode

navigation and commands

Insert mode

typing text

:w

Save

:q!

Exit without saving

:wq

Save and exit

Vim is an editor you launch from the terminal, not a command like ls or cd.

Let's Build Our First Linux Project

```
student@ubuntu: ~  
  
$ pwd  
$ mkdir linux-practice  
$ cd linux-practice  
$ touch app.py  
$ ls  
$ vim app.py  
$ cat app.py  
$ cp app.py backup.py  
$ ls  
$ history
```

Inside Vim

- 1 Press i
- 2 Type:

```
print("Hello Linux")
```
- 3 Press Esc
- 4 Type :wq, press Enter